WEATHER

The weather is the state of the atmosphere with reference to wind, temperature, state of the sea, cloudiness, precipitation, atmospheric pressure, humidity, mist, fog and ice conditions. It is important for seamen to understand all the phenomena connected with the weather and to be able to read weather maps and listen to the weather forecast and report weather conditions at sea such as the visibility and the direction of wind and its force according to the Beaufort Wind Scale.

WINDS

Winds are mainly caused by a difference of temperature which in turn is sometimes responsible for the differences of barometric pressure. The strength and speed of wind at any given time depend on the gradient of atmospheric pressure that is the rate at which pressure changes with distance.

Speed of movement of pressure systems

Slowly:	Moving at less than 15 knots
Steadily:	Moving at 15 to 25 knots
Rather quickly:	Moving at 25 to 35 knots
Rapidly:	Moving at 35 to 45 knots
Very rapidly:	Moving at more than 45 knots

Timing of gale warnings

Imminent:	Within 6 hours of time of issue
Soon:	Within 6 - 12 hours of time of issue
Later:	More than 12 hours from time of issue

Terms referring to wind

Backing;	Indicates the changing of wind in the anticlockwise direction, i.e. from W to SW
Becoming cyclonic:	Indicates that there will be considerable changes in wind direction across the path of a depression within the forecast area
Wind direction:	Indicates the direction from which the wind is blowing
Veering:	Indicates the changing of the wind in a clockwise direction, i.e. from SW to W

Variable: Indicates the wind constantly changing the direction from which it blows

WAVES

Waves are primarily caused by the wind and its action on the surface of the water. Their height depends on how long the wind has been blowing and also on the strength of the wind.

Waves formed by the wind blowing locally are termed “sea”. Waves formed by the wind blowing at a distance from the place of observation are termed “swell”.

Some waves result from earthquakes or underwater seaquakes and on approaching shallow water they become abnormally high and begin to break with great violence causing enormous devastation and loss of life. They are termed “tsunami” and we will all remember the tragic waves caused by a seaquake near Sumatra on Dec. 26th, 2004, which claimed the lives of nearly 300,000 people in South - East Asia.

The following terms are frequently used in connection with waves:

- the length of a wave , that is the horizontal distance from crest to crest or trough to trough. If the distances between the crests of waves are far apart, the sea is termed “a long sea”. When the crests are close together the sea is termed “a short sea”, like for example in the Baltic Sea.
- the height of a wave, that is the vertical distance from trough to crest.
- the period of a wave, that is the time between the passages of two successive wave crests or troughs past a fixed point.
- the velocity of a wave, that is the rate at which the crest travels.

DOUGLAS SEA SCALE

Number	State of sea	Length of swell in metres	Height of swell in metres
0	Calm	-	-
1	Smooth	0 - 5	0 - 0.25
2	Slight	5 - 25	0.25 - 1
3	Moderate	25 - 50	0.75 - 2
4	Rough	50 - 75	2 - 4
5	Very rough	75 - 100	3 - 6

6	High	100 - 135	5 - 7
7	Very high	135 - 200	7 - 10
8	Precipitous	200 - 250	10 - 15
9	Confused	250 - 300	

VISIBILITY

Visibility at sea may be affected by various weather conditions in different parts of the world. In the north it may be affected by rain, sleet, snow, hail and blizzards or snowstorms. In the south it may be affected by torrential rains, drizzle or showers as well as by sand storms. Mist, haze and fog may appear in all areas of the world at different times of the year. The passage of very cold air over much warmer water causes arctic sea smoke, frost smoke or steam fog. It is formed when the lowest layers of the cold air heated by contact with the warm sea tend to rise and are chilled to their dew point on meeting colder air than themselves.

VISIBILITY SCALE

Number	Kind of fog and visibility
0	Dense fog, visibility less than 200 metres
1	Thick fog, visibility from 200 to 300 metres
2	Fog, visibility from 300 to 400 metres
3	Moderate fog, visibility from 400 to 500 metres
4	Thin fog or mist, visibility from 500 - 1000 metres
5	Visibility poor, visibility from 1000 metres to 2 Nm
6	Visibility moderate, visibility from 2 to 5 Nm
7	Visibility good, visibility from 5 to 11 Nm
8	Visibility very good, visibility from 11 to 27 Nm
9	Visibility exceptional, visibility over 27 Nm

CLOUDS AND WEATHER SYMBOLS

Clouds consist of minute drops of water or ice crystals formed by the condensation of water vapour and held in suspension in the atmosphere. There are two main types of clouds: stratiform or layer cloud, resembling fog but not resting on the ground, and cumuliform or white cotton-wool cloud with much greater vertical development than horizontal extent. There are also combinations of these types depending on the height of occurrence and then we speak about cirrus clouds and cirro-cumulus, and cirro-stratus, which are high clouds; alto-cumulus, alto-stratus and nimbo-stratus, which are medium height clouds and strato-cumulus, stratus, cumulus and cumulo-nimbus, which are low clouds.

Clouds usually help in forecasting the weather. Generally speaking, soft round clouds mean fine dry weather with some wind but not very strong. Harsh and jagged clouds mean strong winds. Black clouds mean rain squalls. High clouds moving in a different direction from lower ones foretell a change of the wind.

TYPES OF CLOUDS

Height group	Mean heights in Km	Number	Definition of clouds	Abbreviation	
A High	5 - 13	1 2 3	Cirrus Cirro-cumulus Cirro-stratus	Ci Cc Cs	
B Middle	2 - 7	4 5 6	Alto-cumulus Alto-stratus Nimbo-stratus	Ac As Ns	
C Low	0 - 1	7 8 9 10	Strato-cumulus Stratus Cumulus Cumulo-nimbus	Sc St Cu Cb	

A sailor's experience is also important in foretelling the weather but nowadays seamen rely on weather reports received in a form of facsimile or weather forecasts which are broadcast daily by various stations. In the northern regions and in the Antarctic knowledge of ice terminology is important.

NAVTEX

Navtex is used for passing navigational warnings and meteorological information to ships within the range of 400 Nautical miles off shore. The messages are sent automatically and the information is updated and corrected all the time. Every Navtex message is preceded by a four-letter heading. The first letter characterizes the transmitting station, the second refers to the class of the message, the third and the fourth show the number of received messages in succession. The messages are transmitted on the frequency of 518 kHz. Below is an example of a message transmitted by Navtex.

HIGH SEAS FORECAST

NATIONAL WEATHER SERVICE WASHINGTON DC/TPC MIAMI FLORIDA
MARINE PREDICTION CENTRE/MFB 1700 UTC JAN 7 2002
SUPERSEDED BY NEXT ISSUANCE IN 6 HOURS

SECURITE
PACIFIC NORTH OF 30N AND E OF A LINE FROM BEARING STRAIT TO 50N 160E.
SYNOPSIS VALID 1200 UTC JAN 7. FORECAST VALID 0000 UTC JAN 9.

WARNINGS
STORM 45N 175W 953MB WILL MOVE NE 25KT. WINDS 45 TO 60KT SEAS 25 TO 40 FT ELSEWHERE WITHIN 660NM S AND 240NM N SEMICIRCLES.
ALSO WINDS 30 TO 45 KT SEAS 15 TO 25 FT ELSEWHERE FROM 30N TO 54N… BETWEEN 155W AND 160E. FORECAST COMPLEX STORM 54N 156W 957MB.
FORECAST ASSOCIATED COLD FRONT THROUGH 55N 150W AND 46N 146W.
FORECAST WINDS 40 TO 55 KT AND SEAS 18 TO 28 FT WITHIN 300 NM E OF THE FRONT.
GALE 42N 142W 986 MB WILL MOVE NE 30 KT. WINDS 30 TO 45 KT AND SEAS 15 TO 25 FT WITHIN 720 NM SE SEMICIRCLE AND 480 NM W QUADRANT. FORECAST DISSIPATED INLAND. FORECAST CONDITIONS AS DESCRIBED ABOVE.
AREAS OF MODERATE TO HEAVY FREEZING SPRAY OVER THE BEARING SEA N OF 55N W OF 160W. FORECAST AREAS OF MODERATE TO FREEZING SPRAY IN BEARING SEA N OF 52N W OF 162W.
SYNOPISIS AND FORECAST
FORECAST S WINDS 20 TO 30 KT FROM 35 KT TO 45 KT BETWEEN 160E TO 162E.